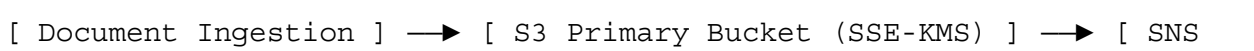
6th lab: AWS Enterprise Hybrid Security & Asynchronous Processing Pipeline

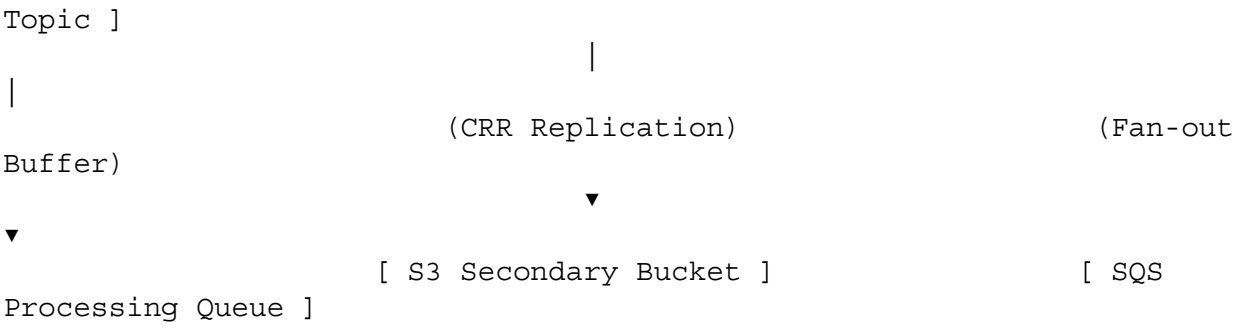
Estimated Duration: 1.5 to 2 Hours
Focus Areas: AWS KMS (Envelope Encryption), Amazon S3 (Cross-Region Replication & Key Policies), AWS KMS Grant/Policy Delegation, Amazon SNS & SQS Event Notifications, Least-Privilege IAM Policies, Enterprise Compliance & Auditing.

Architecture Overview & Blueprint

This hands-on lab introduces a critical enterprise scenario frequently tested on the AWS Solutions Architect Associate (SAA-C03) exam: **asynchronous, multi-region event-driven processing combined with end-to-end customer-managed key (CMK) envelope encryption.**

Component	AWS Service	Architectural Purpose
Ingestion Tier	Amazon S3 (Primary Region)	Ingests enterprise documents, encrypted at rest using AWS KMS (SSE-KMS).
Messaging Layer	Amazon SNS & Amazon SQS	Publishes S3 create events to an SNS topic, fan-out buffering to SQS queues for asynchronous processing.
Cross-Region DR Tier	Amazon S3 (Secondary Region)	Stores cross-region replicated (CRR) encrypted backups for compliance and disaster recovery.
Key Management Tier	AWS KMS (Primary & Secondary)	Provides envelope encryption across both regions with customized key policies permitting cross-service usage.
Audit Tier	AWS CloudTrail & CloudWatch	Captures data-plane API actions (kms:GenerateDataKey, s3:GetObject) for real-time compliance auditing.





Description of the Project

In this lab, you will design and implement a zero-trust compliance ingestion architecture for an enterprise client.

1. **Multi-Region Disaster Recovery:** You will provision two private S3 buckets across two distinct AWS regions (e.g., us-east-1 and us-west-2) and configure **S3 Cross-Region Replication (CRR)** using custom customer-managed KMS keys in both regions.
2. **KMS Policy & Delegation:** You will write strict KMS Key Policies allowing S3 and SQS to perform envelope encryption operations (kms:GenerateDataKey, kms:Decrypt) without granting overly permissive broad access.
3. **Event Notification Fan-Out:** You will configure S3 Event Notifications to send messages to an **Amazon SNS Topic**, which fans out to an **Amazon SQS Queue**, demonstrating decoupled asynchronous notification patterns.
4. **Security & Validation:** You will test key permissions, intentionally trigger KMS Access Denied errors, inspect CloudTrail event logs, and verify that cross-region replication succeeds with KMS re-encryption at the destination.

Step-by-Step Implementation Guide

Step 1: Create Primary and Secondary KMS Customer Managed Keys (CMKs)

To maintain compliance, data in both regions must be encrypted using dedicated, region-specific KMS keys.

- Navigate to **AWS KMS Console** in **Region 1** (us-east-1):
  - Click **Create Key**.
  - **Key Type:** Symmetric | **Key Usage:** Encrypt and Decrypt.
  - **Alias:** alias/primary-document-kms-key.
  - Define Administrative and Usage permissions (keep defaults for current IAM user/role).
- Navigate to **AWS KMS Console** in **Region 2** (us-west-2):
  - Click **Create Key**.
  - **Key Type:** Symmetric | **Key Usage:** Encrypt and Decrypt.
  - **Alias:** alias/secondary-document-kms-key.
  - Define Administrative and Usage permissions (keep defaults for current IAM

user/role).
- Copy both **KMS Key ARNs** for use in subsequent steps.

Step 2: Provision S3 Buckets & Configure Cross-Region Replication (CRR)

- **Create Primary Bucket (us-east-1):**
  - **Bucket Name:** enterprise-primary-docs-<unique-id>
  - **Versioning:** Enable (Required for Replication).
  - **Default Encryption:** Server-side encryption with AWS Key Management Service keys (SSE-KMS) -> Choose alias/primary-document-kms-key.
  - Enable **Bucket Key**.
- **Create Secondary Bucket (us-west-2):**
  - **Bucket Name:** enterprise-dr-docs-<unique-id>
  - **Versioning:** Enable.
  - **Default Encryption:** Server-side encryption with AWS Key Management Service keys (SSE-KMS) -> Choose alias/secondary-document-kms-key.
  - Enable **Bucket Key**.
- **Configure IAM Replication Role:**
  - Navigate to **IAM Console ➔ Roles ➔** Click **Create Role**.
  - **Trusted Entity:** AWS Service ➔ Select **S3**.
  - **Role Name:** S3KMSReplicationRole.
  - Attach inline policy to grant S3 read/replica permissions and KMS decryption/encryption rights on both keys:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["s3:GetReplicationConfiguration", "s3:ListBucket"],
      "Resource": "arn:aws:s3:::enterprise-primary-docs-<unique-id>"
    },
    {
      "Effect": "Allow",
      "Action": ["s3:GetObjectVersionForReplication",
"s3:GetObjectVersionAcl"],
      "Resource": "arn:aws:s3:::enterprise-primary-docs-<unique-id>/*"
    },
    {
      "Effect": "Allow",
      "Action": ["s3:ReplicateObject", "s3:ReplicateDelete"],
      "Resource": "arn:aws:s3:::enterprise-dr-docs-<unique-id>/*"
    },
    {
      "Effect": "Allow",
      "Action": ["kms:Decrypt"],
```

```
      "Resource": "<PRIMARY_KMS_KEY_ARN>"
    },
    {
      "Effect": "Allow",
      "Action": ["kms:Encrypt", "kms:GenerateDataKey"],
      "Resource": "<SECONDARY_KMS_KEY_ARN>"
    }
  ]
}
```

- **Update Key Policy on Destination KMS Key (us-west-2):**
  - Add a statement to the destination KMS key policy allowing S3KMSReplicationRole to use the key for kms:Encrypt and kms:GenerateDataKey.
- **Enable Replication Rule:**
  - In Primary Bucket (us-east-1) ➔ **Management** tab ➔ **Replication rules** ➔ Click **Create replication rule**.
  - **Status:** Enabled | **Source:** Apply to all objects in the bucket.
  - **Destination:** Select enterprise-dr-docs-<unique-id> in us-west-2.
  - **IAM Role:** Select S3KMSReplicationRole.
  - **Encryption:** Select *Replicate objects encrypted with AWS KMS*. Specify the destination KMS Key ARN (alias/secondary-document-kms-key).

Step 3: Configure Decoupled Messaging (SNS & SQS) with KMS Protection

- **Create KMS-Encrypted SNS Topic (us-east-1):**
  - Navigate to **Amazon SNS Console ➔ Topics ➔** Click **Create Topic**.
  - **Type:** Standard | **Name:** document-uploaded-topic.
  - **Encryption:** Enable. Select alias/primary-document-kms-key.
- **Create SQS Queue (us-east-1):**
  - Navigate to **Amazon SQS Console ➔ Queues ➔** Click **Create Queue**.
  - **Type:** Standard | **Name:** document-processing-queue.
  - **Encryption:** SSE-KMS with alias/primary-document-kms-key.
- **Subscribe SQS Queue to SNS Topic:**
  - In SQS Console ➔ Select document-processing-queue ➔ Click **Subscribe to Amazon SNS Topic**.
  - Choose document-uploaded-topic.
- **Update Primary KMS Key Policy for S3, SNS, and SQS:**
  - Modify your Primary KMS Key Policy in us-east-1 to permit S3 to publish to SNS and SNS to publish to SQS:

```
{
  "Sid": "AllowS3AndSNSToUseKey",
  "Effect": "Allow",
  "Principal": {
    "Service": ["s3.amazonaws.com", "sns.amazonaws.com",
"sqs.amazonaws.com"]
  },
```

```
  "Action": [
    "kms:GenerateDataKey*",
    "kms:Decrypt"
  ],
  "Resource": "*"
}
```

- **Configure S3 Event Notifications:**
  - Navigate to Primary S3 Bucket ➔ **Properties** ➔ **Event Notifications** ➔ Click **Create Event Notification**.
  - **Event Types:** Select All object create events (s3:ObjectCreated:*).
  - **Destination:** Select SNS Topic ➔ Choose document-uploaded-topic.

Step 4: Testing, Auditing, and Verification

- **Test Ingestion & Fan-Out:**
  - Upload a sample document (e.g., confidential-report.pdf) into the primary S3 bucket (us-east-1).
  - Check the SQS queue (document-processing-queue) ➔ Click **Send and receive messages** ➔ Click **Poll for messages**.
  - Verify that an S3 creation notification payload wrapped inside an SNS message structure is received successfully.
- **Test Cross-Region Replication:**
  - Navigate to the DR S3 bucket in us-west-2 (enterprise-dr-docs-<unique-id>).
  - Verify that confidential-report.pdf appears and its server-side encryption reflects alias/secondary-document-kms-key.
- **Validate Least-Privilege IAM / Access Denied:**
  - Attempt to download the replicated file from us-west-2 using an IAM user/role that lacks kms:Decrypt access to alias/secondary-document-kms-key.
  - Confirm that AWS returns HTTP 403 / AccessDenied despite having S3 s3:GetObject permissions.
- **Inspect CloudTrail Logs:**
  - Open **AWS CloudTrail Console ➔ Event history**.
  - Filter by **Event name:** GenerateDataKey or Decrypt.
  - Verify the KMS data-plane calls made by S3 and S3 Replication services.

Steps performed (Lab Verification Checklist)

I created Primary and Secondary Symmetric KMS Keys

Successfully provisioned symmetric KMS keys alias/primary-document-kms-key in us-east-1 and alias/secondary-document-kms-key in us-west-2.

I created Primary and DR S3 Buckets

Configured versioning and default KMS encryption on primary and destination S3 buckets for

Cross-Region Replication.

Created IAM Role S3KMSReplicationRole

Attached inline policies granting cross-region S3 replication permissions and KMS decryption/encryption authorization across both regions.

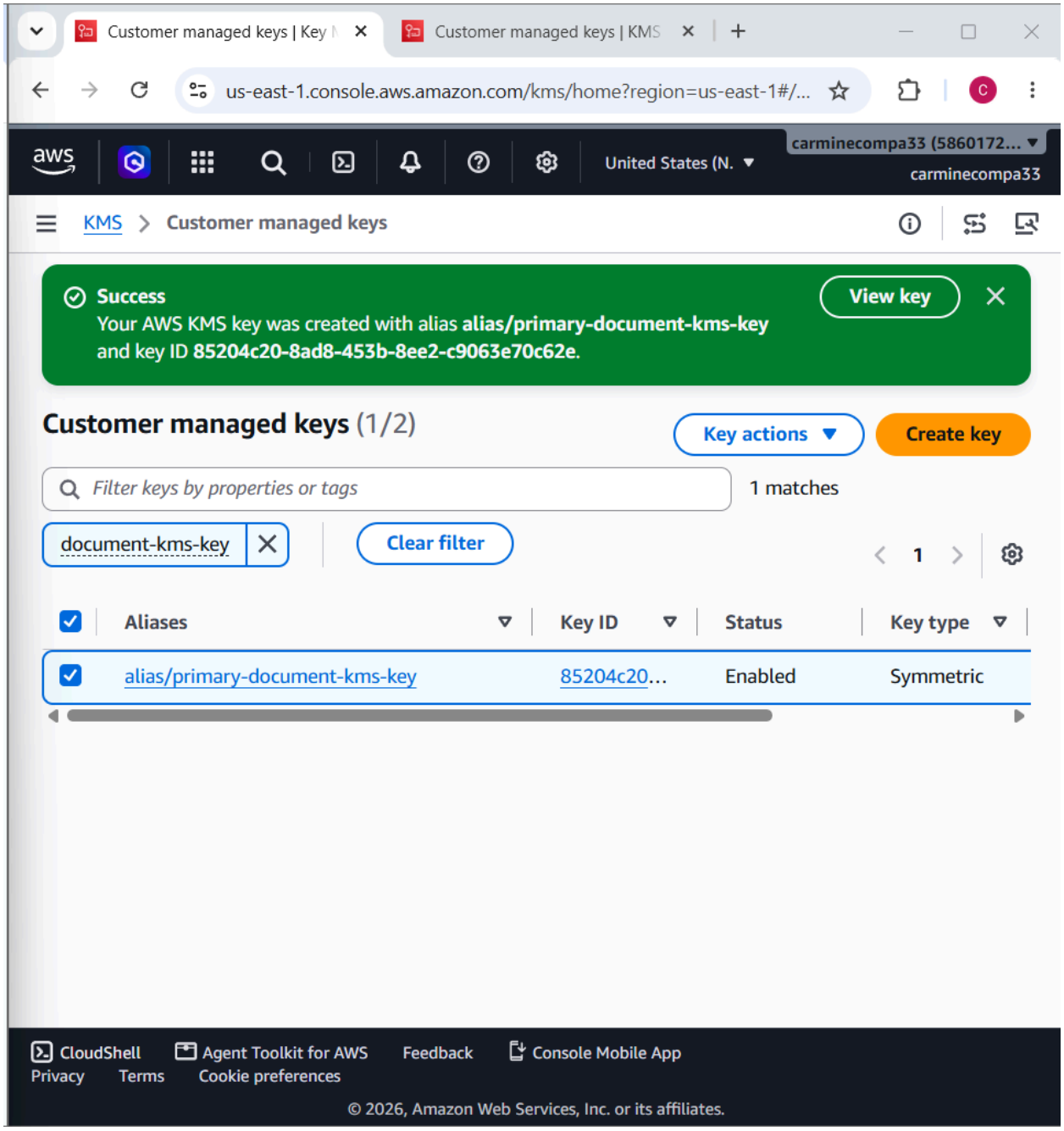
Configured KMS Key Policy for S3, SNS, and SQS

Delegated access to AWS managed services to permit encrypted event fan-out from S3 through SNS to SQS queues.

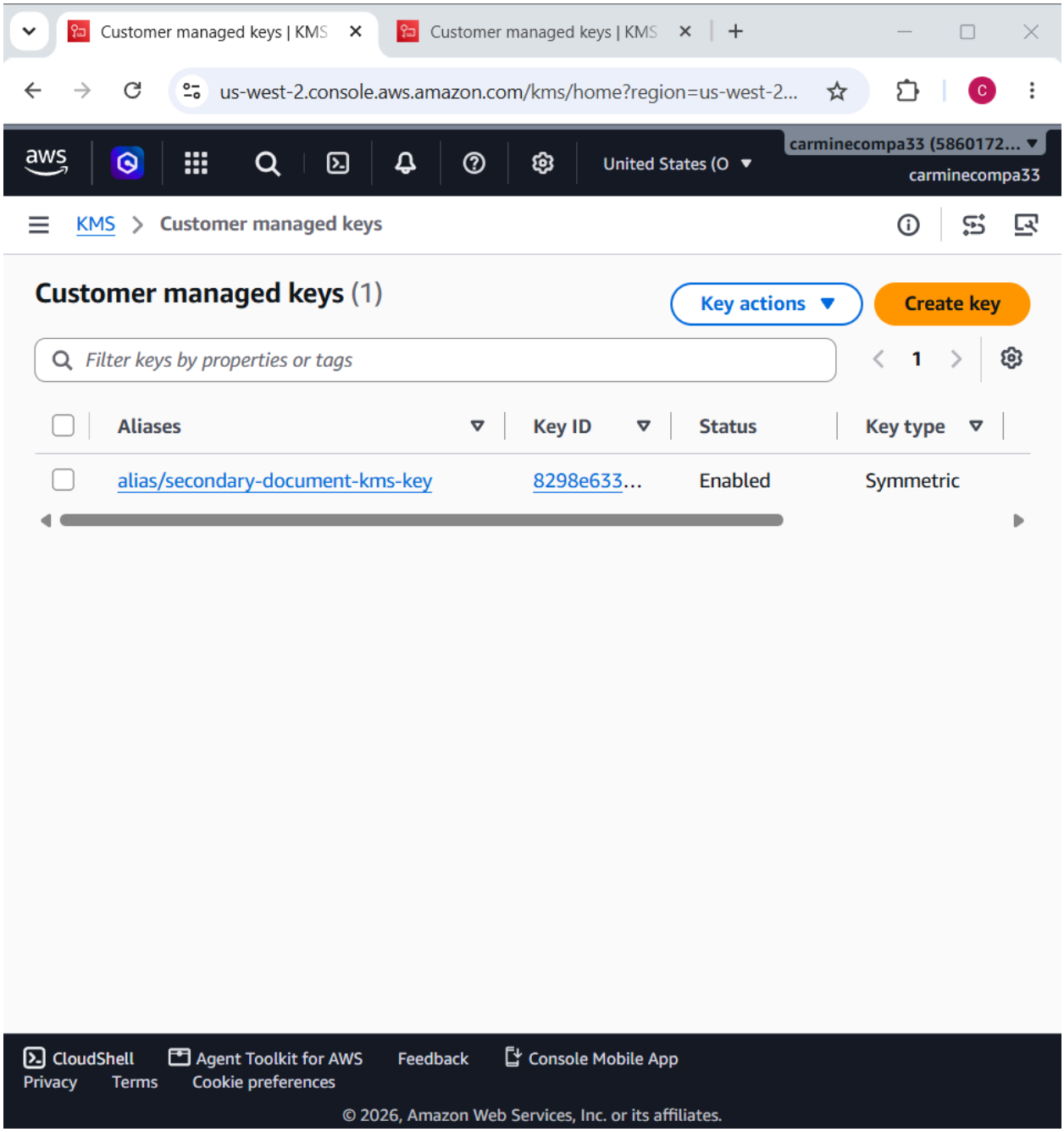
Verified End-to-End Pipeline and DR Replication

Confirmed automated message delivery into SQS upon S3 upload, verified object replication with target KMS re-encryption, and audited access logs via CloudTrail.

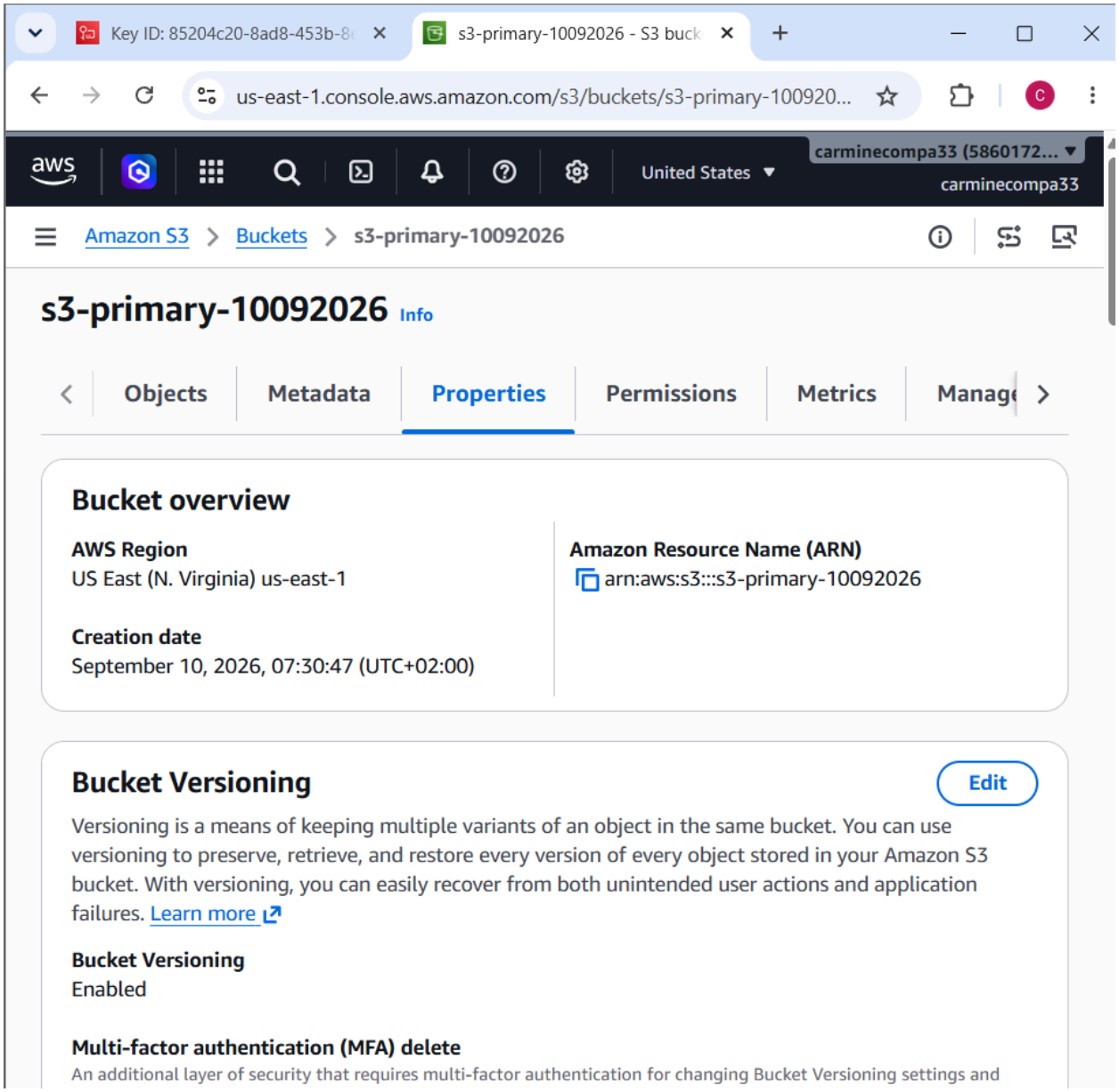
Created customer managed key in us-east-1 (kms-encryption)



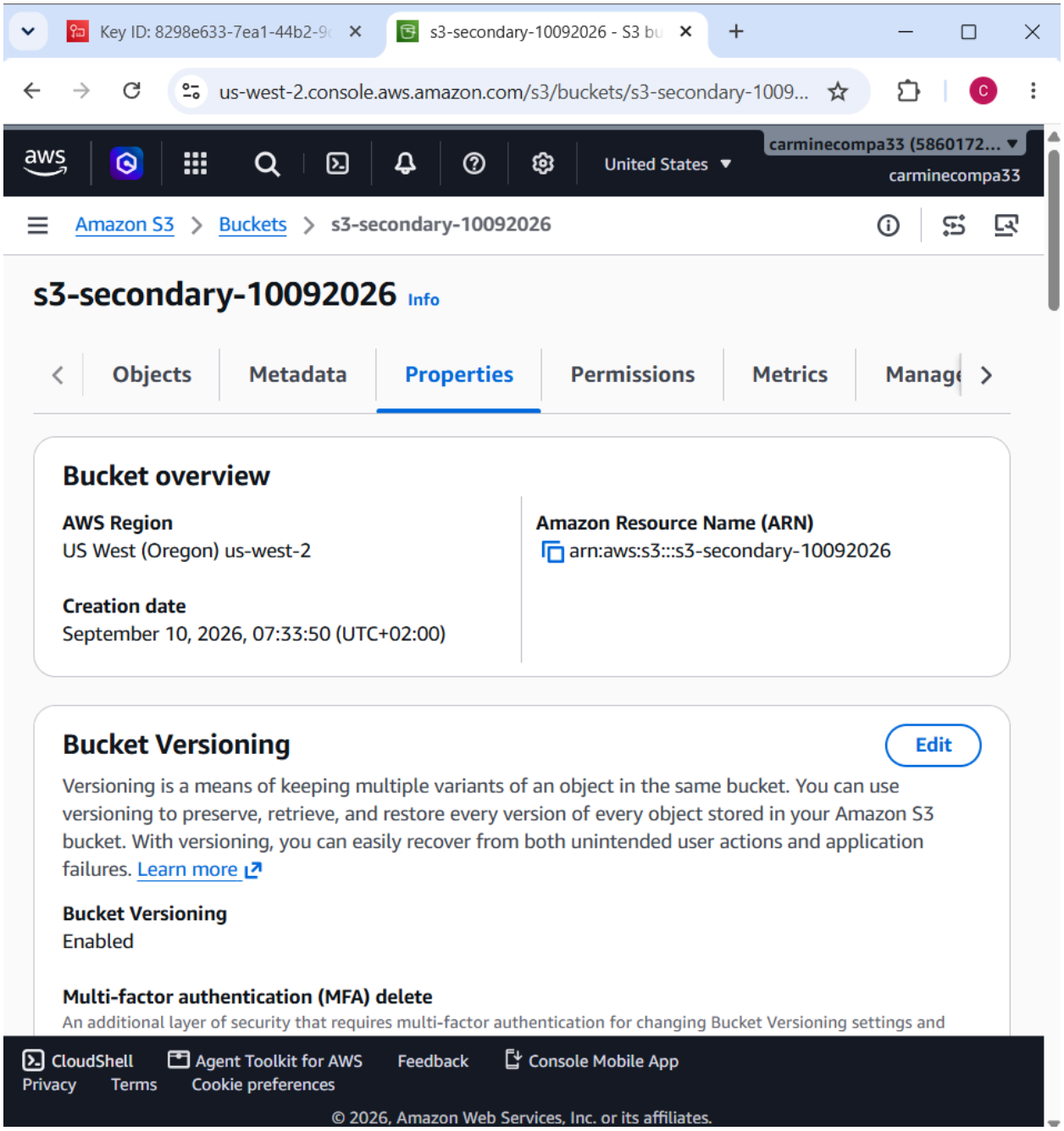
Created customer managed key in us-west-2



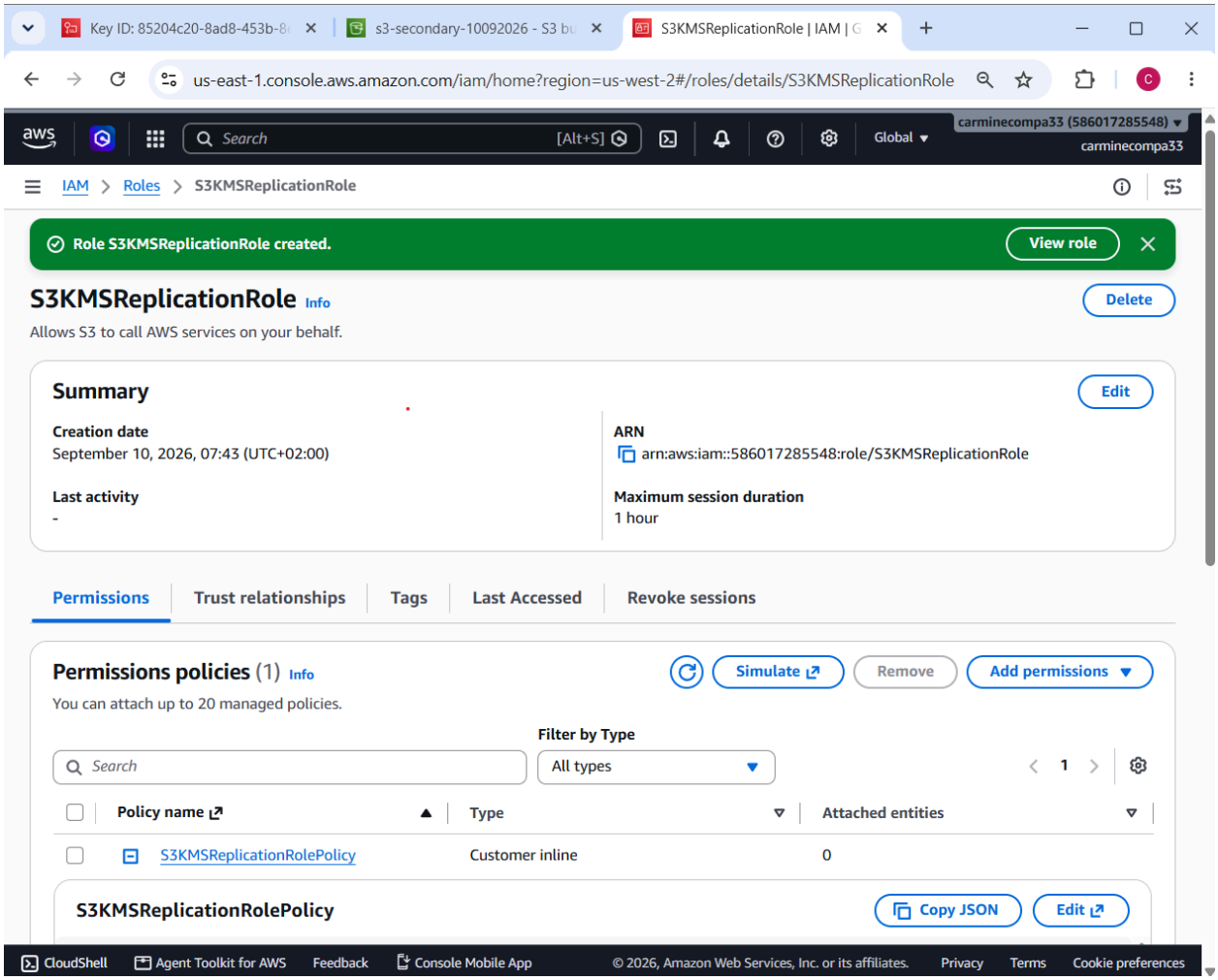
Created S3 bucket in eu-east-1 with versioning and key enabled (kms encryption)



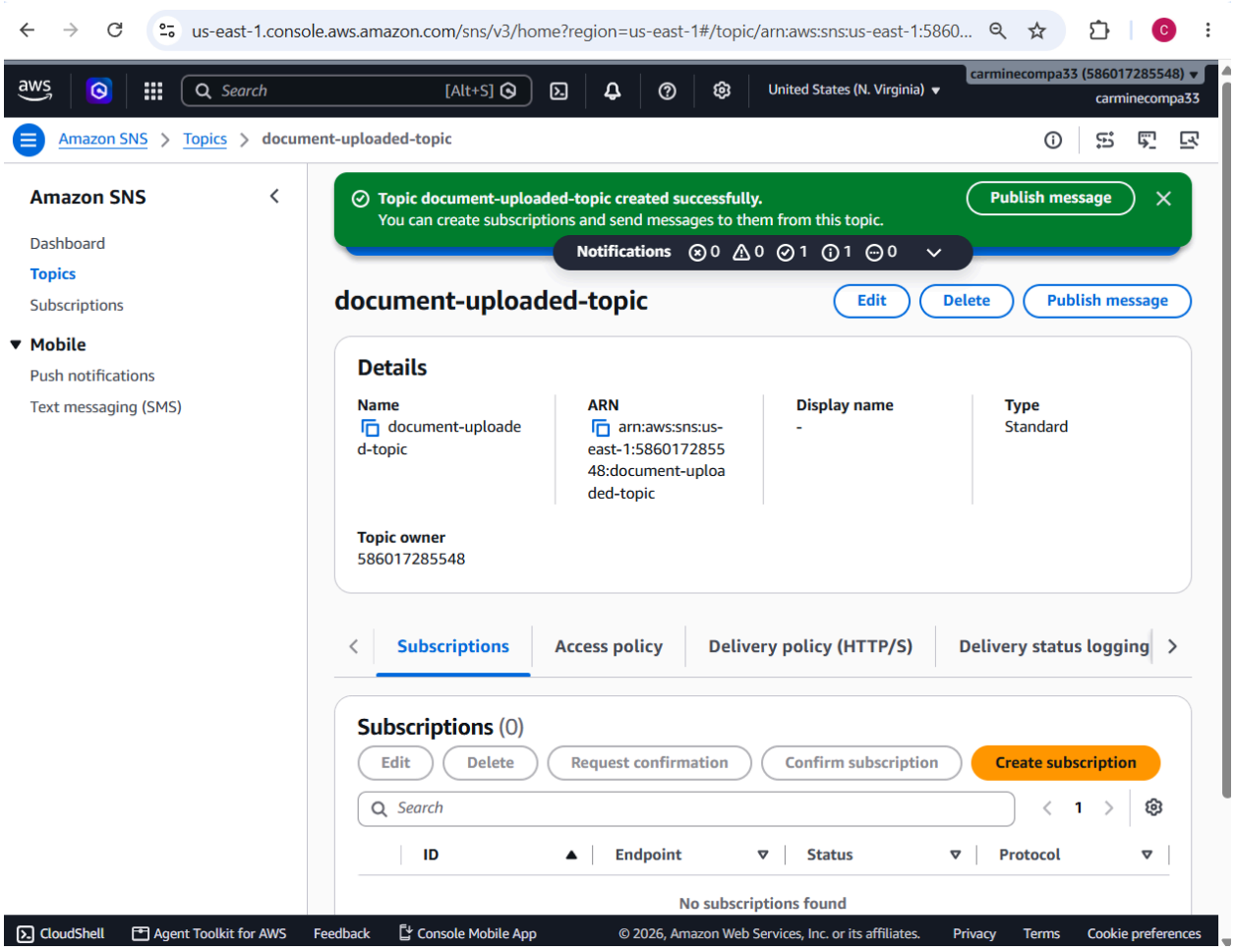
Created s3 secondary bucket in us-west-2 (with version and key enabled)



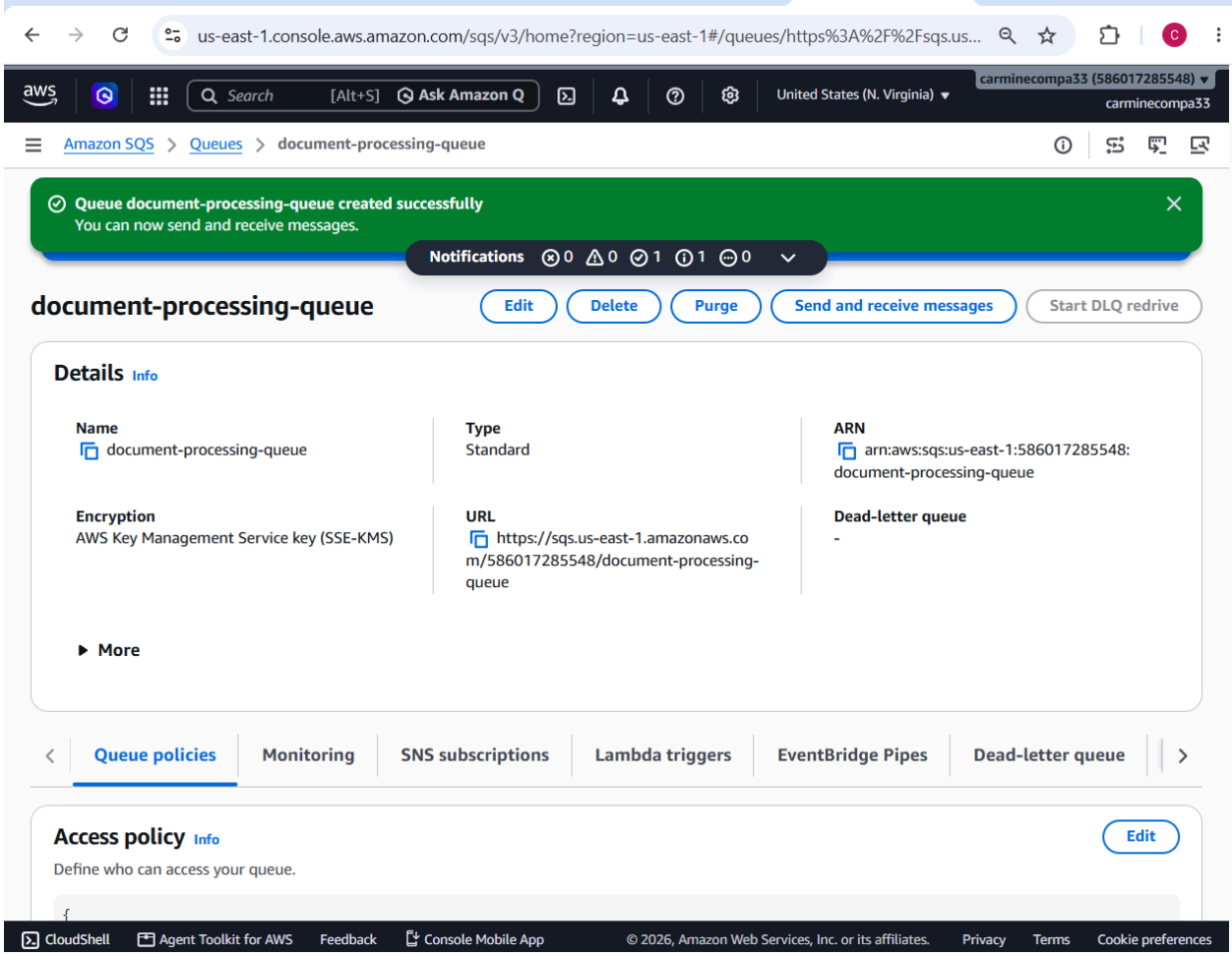
Created S3KMSReplicationRole



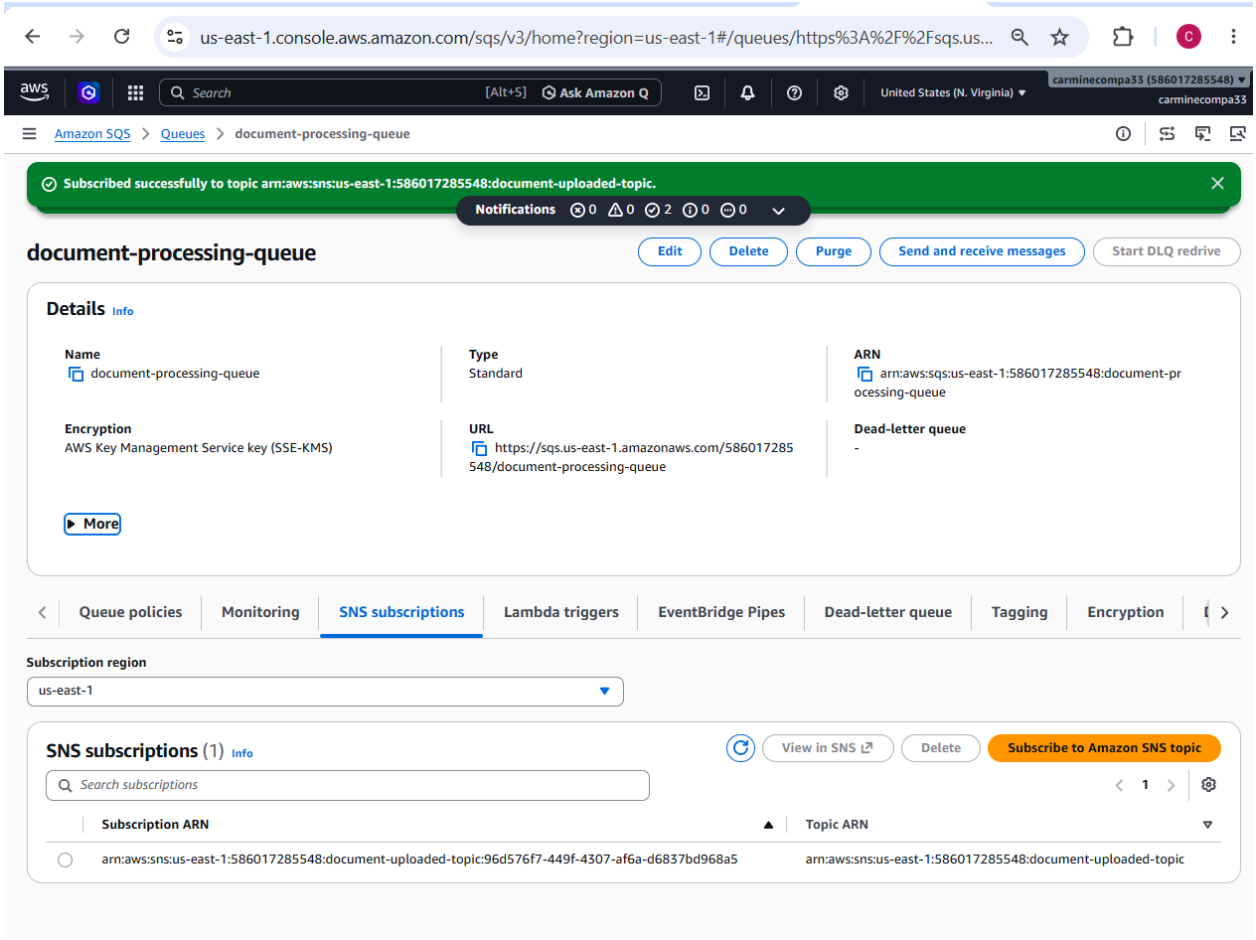
Create Simple Notification Service topic with customer managed key encryption



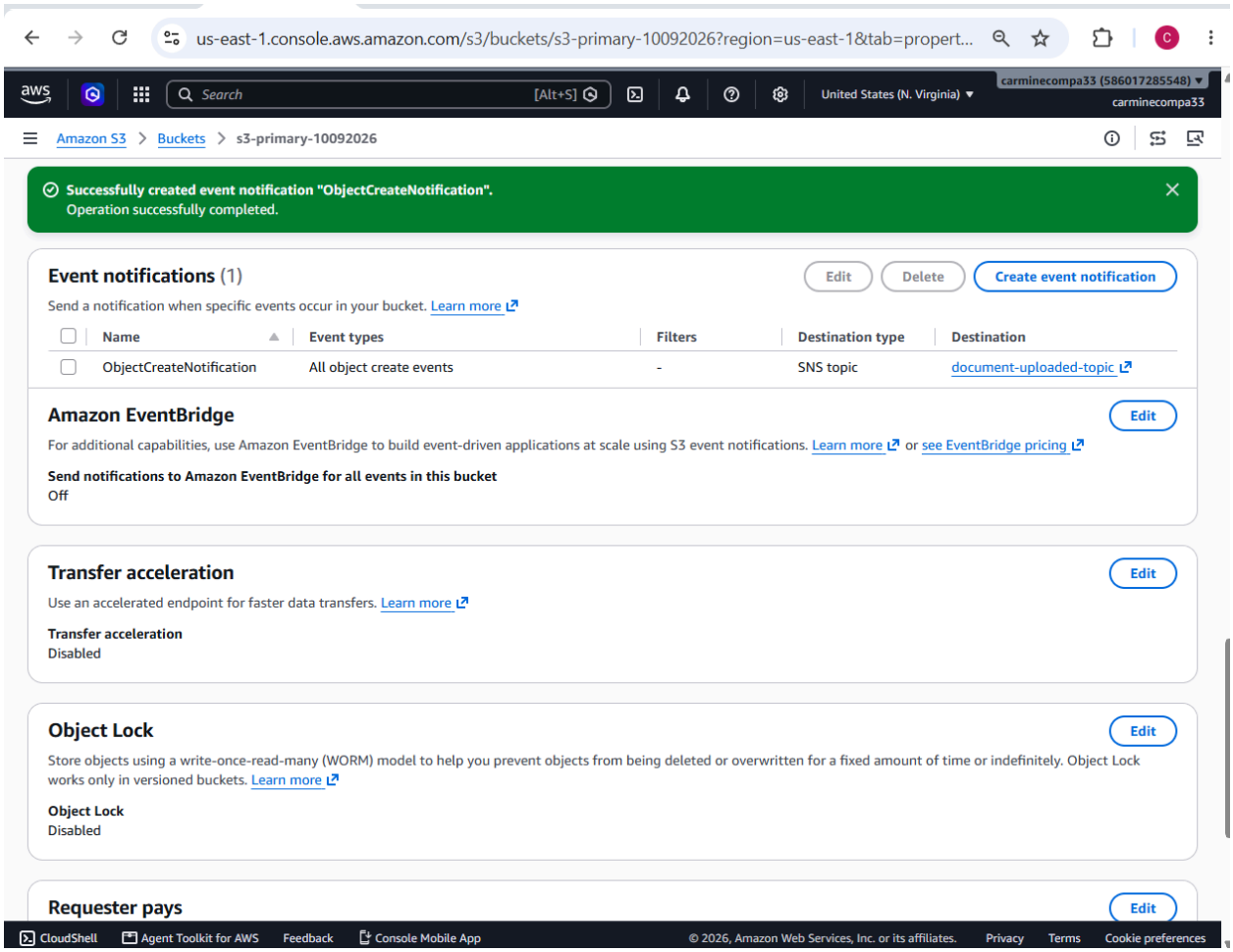
Created Simple Queue Service in us-east-1 with customer managed key encryption



Subscribed to the Simple Notification Service (SNS) topic from the Simple Queue Service (SQS)

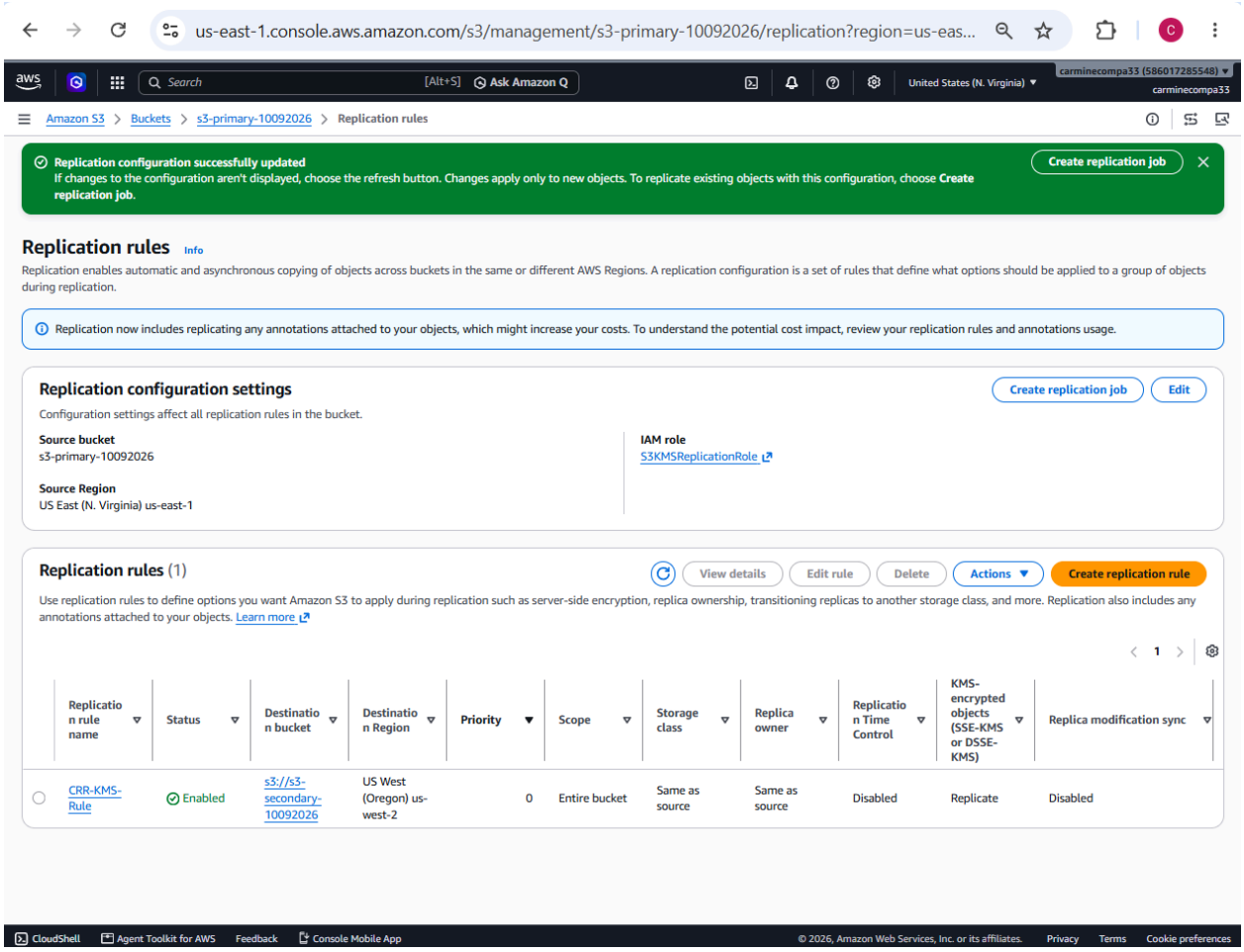


Created event notification in S3 for ObjectCreateNotification with the SNS topic as destination

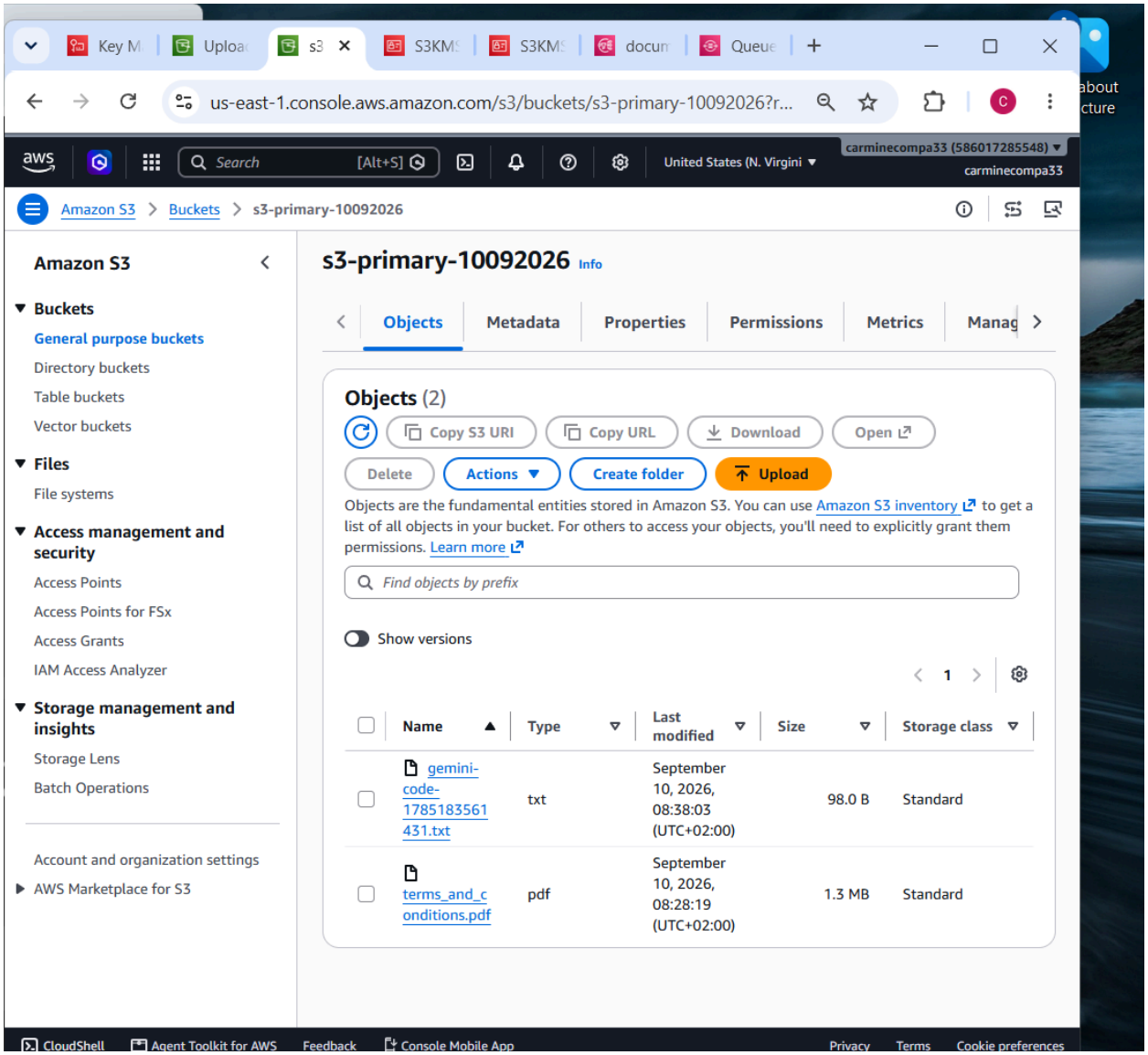


Create replication rule in s3 primary bucket (us-east-1) to s3 secondary bucket (us-west-2) with

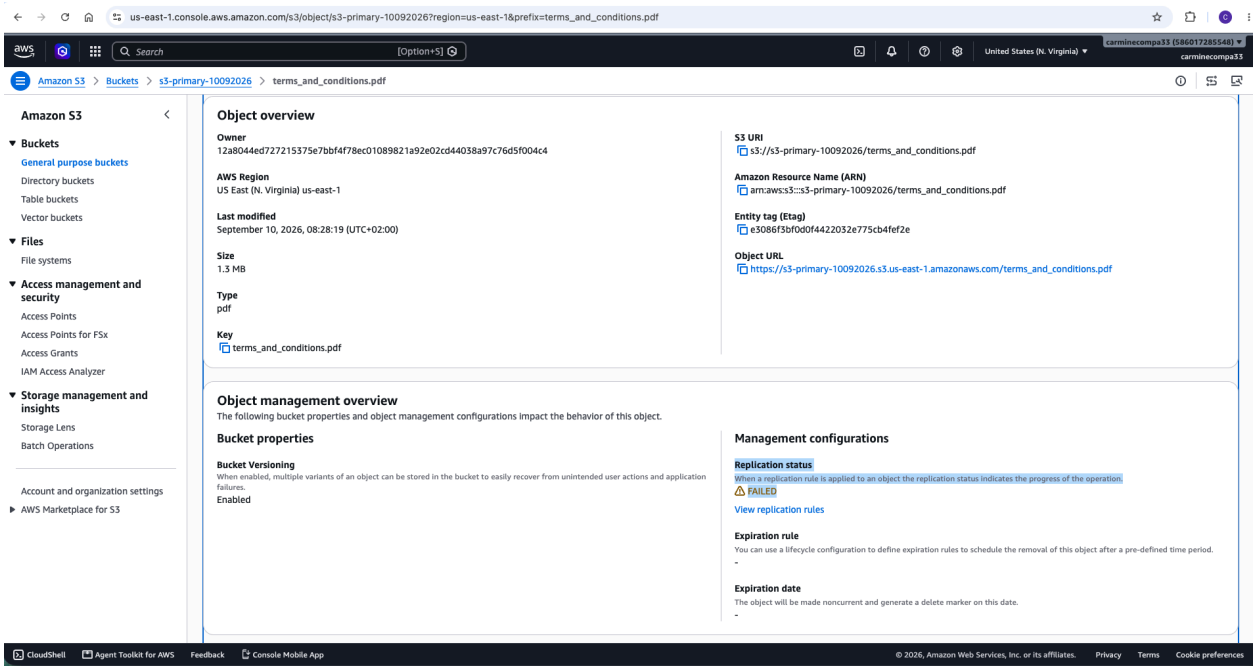
encryption with customer managed key us-west-2



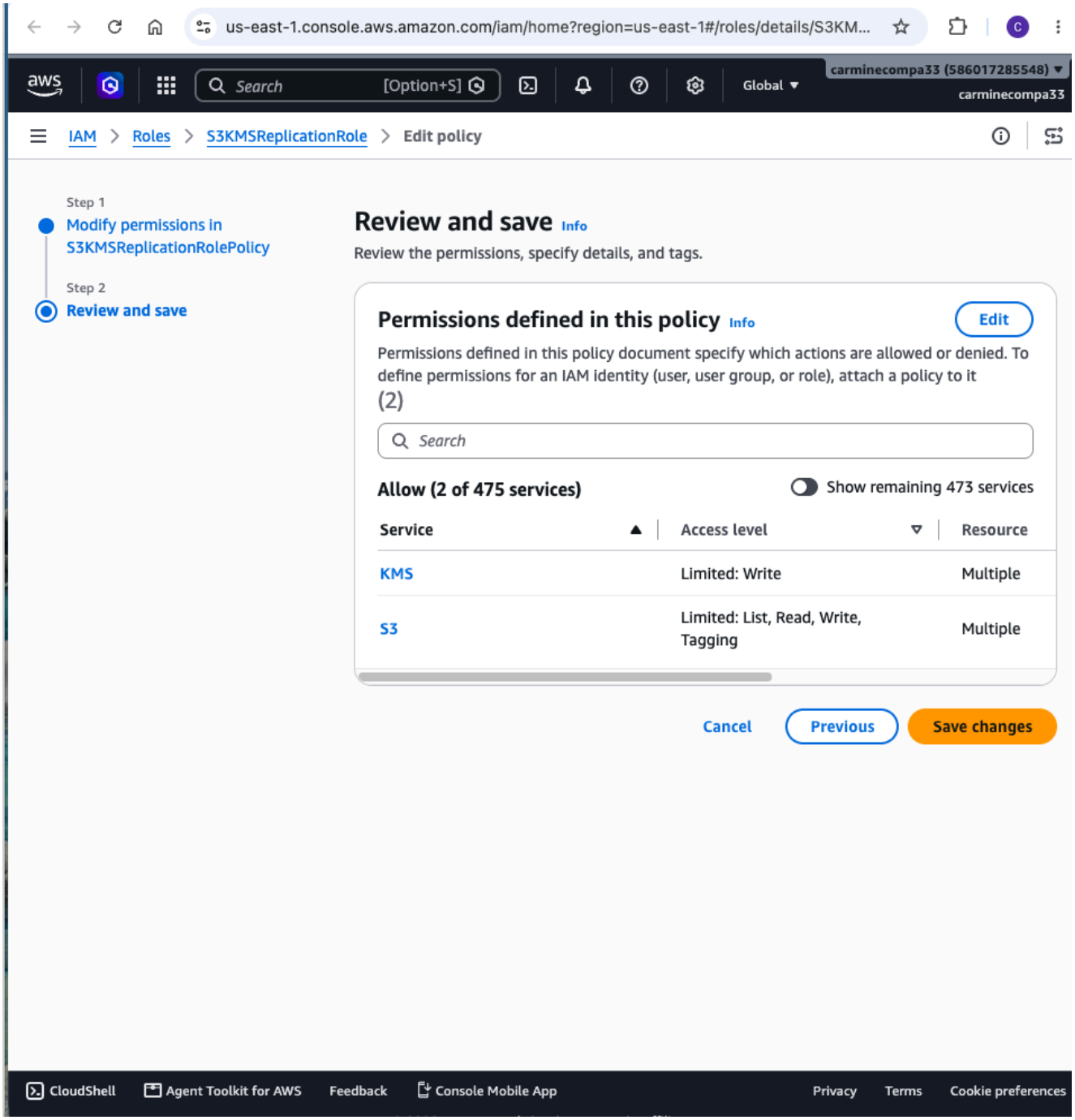
I upload a file in S3 primary bucket in us-east-1



However the file did not get automatically replicated in S3 secondary bucket in us-west-2:

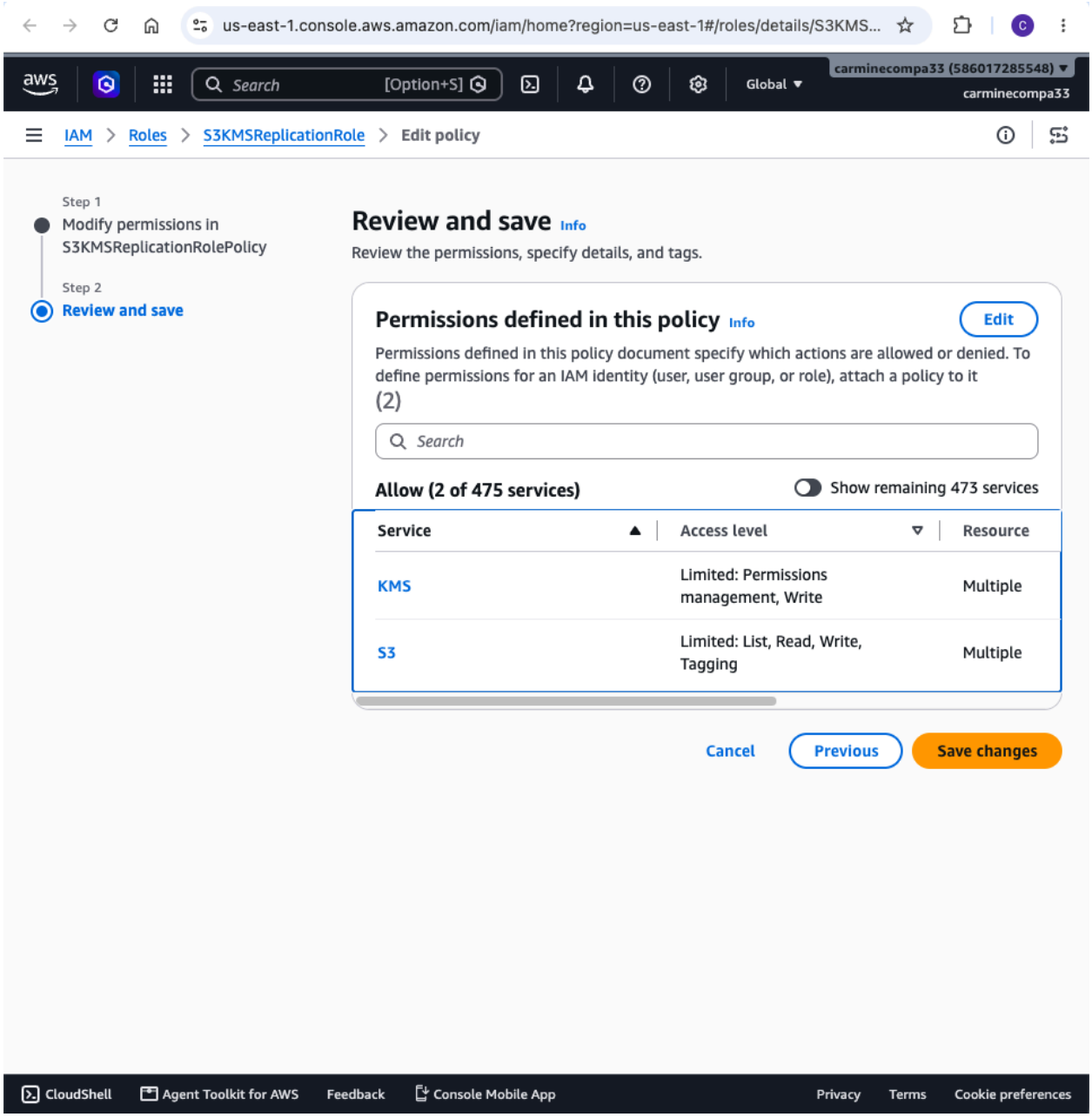


Let’s modify IAM role inline policy

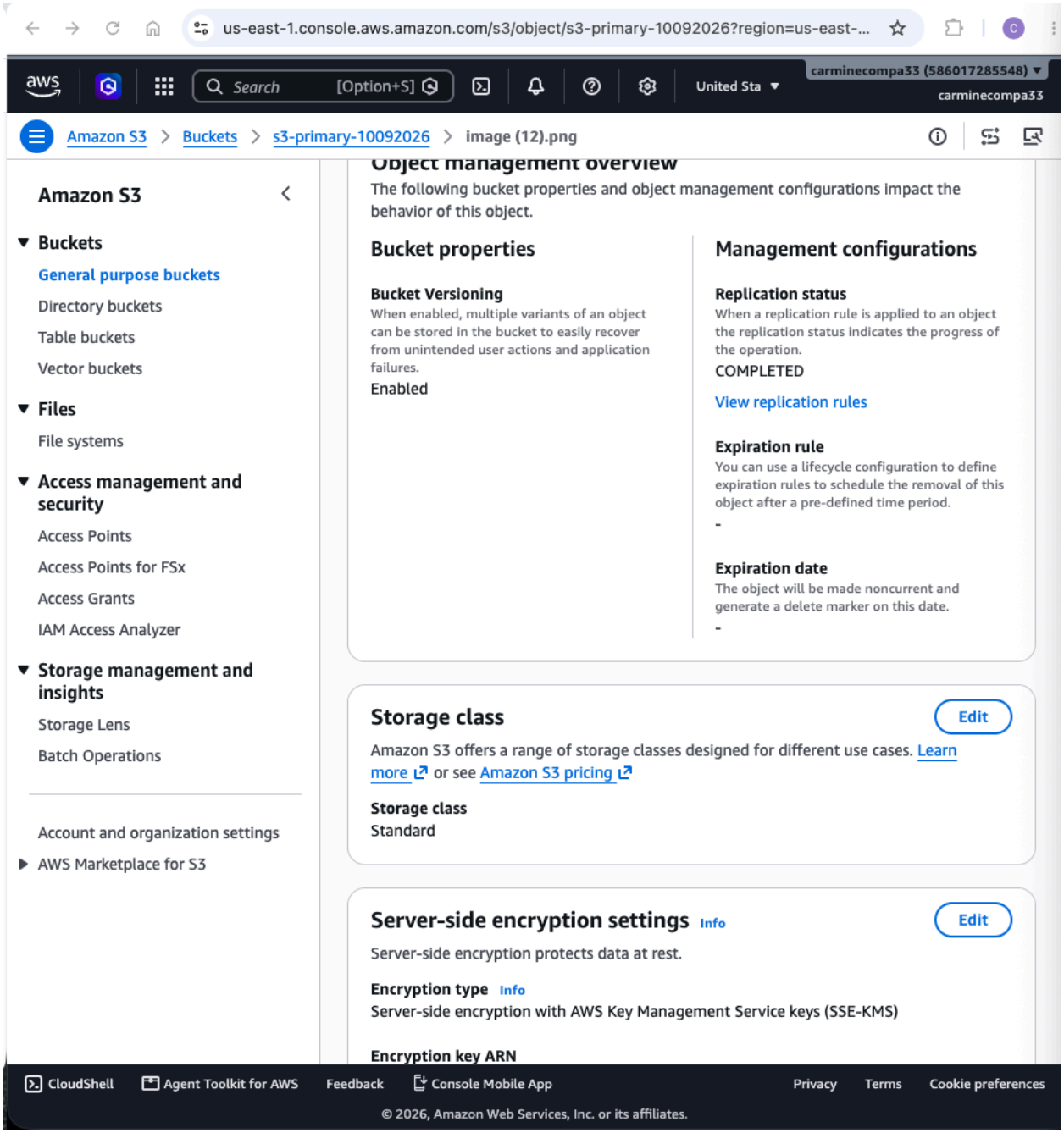


Uploaded a new file but it still fails

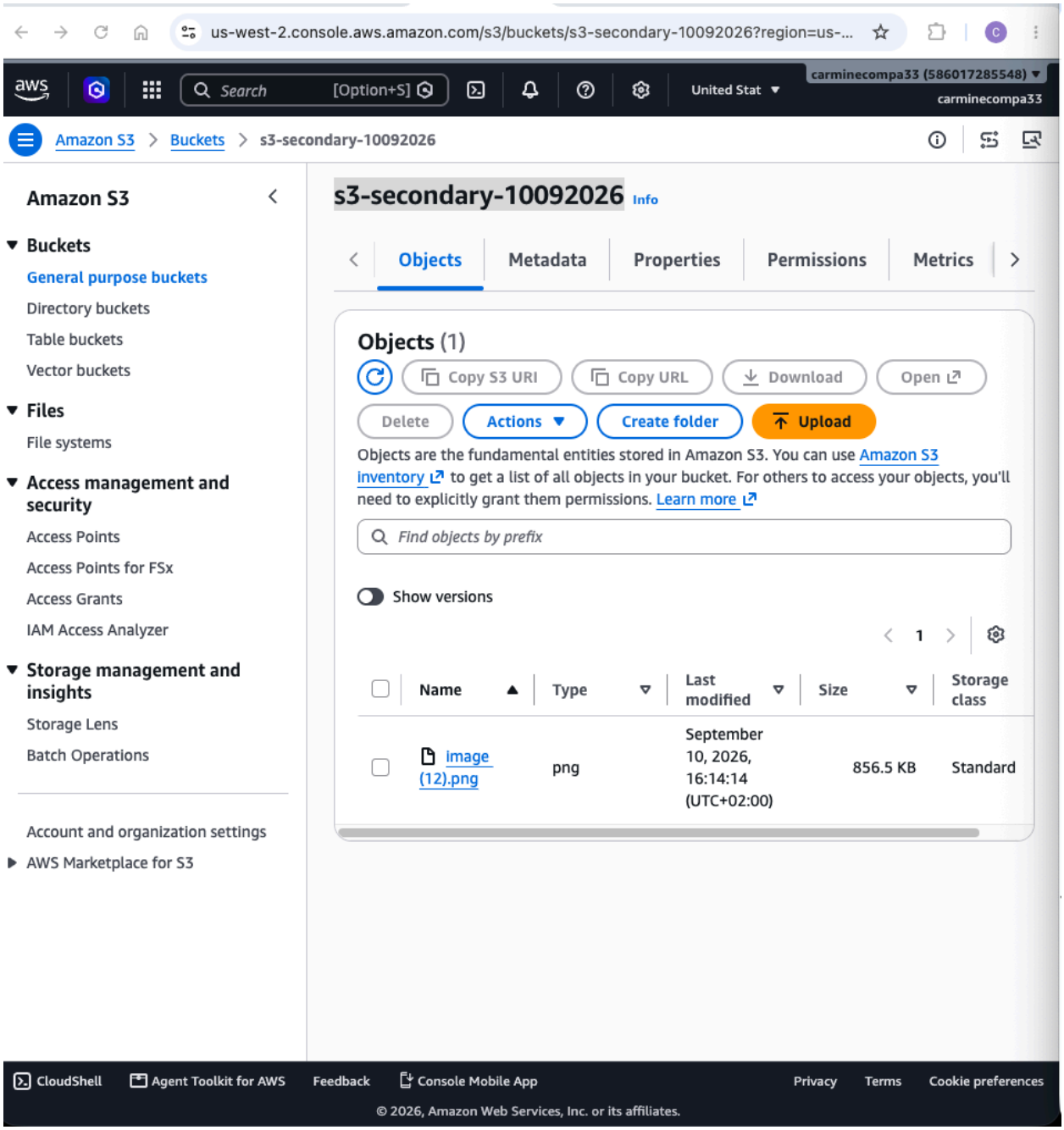
Let’s modify again S3KMSReplicationRole



Now Replication status is COMPLETED on the file uploaded in s3 primary:



And the file is present in s3-seconday:


us-west-2.console.aws.amazon.com/s3/buckets/s3-secondary-10092026?region=us-...
Search
[Option+S]
United Stat
carminecompa33 (586017285548)
carminecompa33
Amazon S3
Buckets
s3-secondary-10092026
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization settings
AWS Marketplace for S3
s3-secondary-10092026
Info
Objects
Metadata
Properties
Permissions
Metrics
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Show versions
1
Name
Type
Last modified
Size
Storage class
image (12).png
png
September 10, 2026, 16:14:14 (UTC+02:00)
856.5 KB
Standard
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
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